PYL1002: Waves and Oscillations | **Tutorial - 10** (The Last One)

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Submit on Moodle by Sunday, 19/4/2026 midnight

Interfering with Interference

1. This tutorial is a "take-home" assignment. Submissions will be done by each student on Moodle by midnight, Sunday, April 19, 2026.
2. Each student will work on this tutorial. You are allowed to collaborate with your friends and classmates. If you do so, mention their names on top of your submission. Credit where credit is due.
3. Each student should submit a single pdf as their submission on Moodle. A reasonable attempt of this assignment will earn you tutorial attendance for Saturday, April 18 (which is functioning as a Tuesday).
4. Use of any smart device or AI-assistance is not allowed unless explicitly stated in the question. You are bound by the Honor Code of the class.

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1 A Bit More on N-Slit Interference

In class, we studied the N-slit interference pattern with distance d between successive slits,

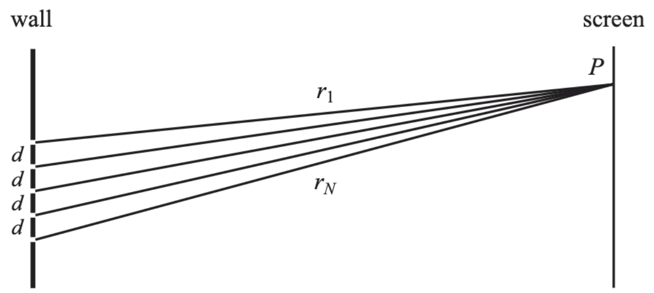


Figure 1: Schematic of N-slit interference, here shown for $N = 5$ slits. Credit: Morin

and found the amplitude ratio relative to the central maxima to be,

$$\frac{A_{\rm tot}(\theta)}{A_{\rm tot}(0)} = \frac{\sin\left(N\alpha/2\right)}{N\sin\left(\alpha/2\right)}\,, \tag{1}$$

where $\alpha = kd\sin\theta$. This was the result in the *far-field* limit and looked at small angles where we can ignore the decay of the amplitude due to cylindrical nature of each wavefront.

(a) Show that in the case of the Young's Double Slit Experiment with $N = 2$, this reduces to the standard result we derived in class (in the small angle $\theta \ll 1$ approximation[1]),

$$\frac{I_{\rm tot}(\theta)}{I_{\rm tot}(0)} \approx \cos^2\left(\frac{\pi d\sin\theta}{\lambda}\right) \tag{2}$$

(b) Next, for this $N = 2$ result of Eq. (2), explicitly find the angular locations of the maximas and minimas. Verify that the result is indeed what you learned (and hopefully remember) from your JEE days.

(c) Back to the N-slit case. I claimed in class that the $I_{\rm tot}(\alpha)/I_{\rm tot}(0)$ result is periodic in α with a period of 2π. Quickly verify this.

(d) Draw the pattern $I_{\rm tot}(\alpha)/I_{\rm tot}(0)$ as a function of α for $N = 6$. Clearly mark the location of all the minimas and primary maximas of the pattern.

(e) In general, for a given N, how many little bumps (small maximas) are in between any two primary maximas?

[1] You will notice this result misses the $\cos\theta$ term we had in class for Young's Souble Slit. This term reflects the amplitude decay with distance which we are ignoring here due to small angle approximation for convenience.

2 LIDARs and Autonomous Cars

In class, we discussed how autonomous cars use interference as part of their LIDAR (Light Detection and Ranging) sensors to map their surroundings. In particular, a set of N-emitters (lasers, for example) have the following interference pattern, and the primary maximas can be thought of as the direction of the "focussed" beams sent out from the LIDAR for mapping. The intensity pattern (normalised to the central maximum) is

$$\frac{I(\alpha)}{I(0)} = \frac{1}{N^2}\left(\frac{\sin(N\alpha/2)}{\sin(\alpha/2)}\right)^2, \qquad \alpha = kd\sin\theta = \frac{2\pi d\sin\theta}{\lambda}. \tag{3}$$

Here is the schematic for $N = 8$ emitters from a LIDAR sensor:

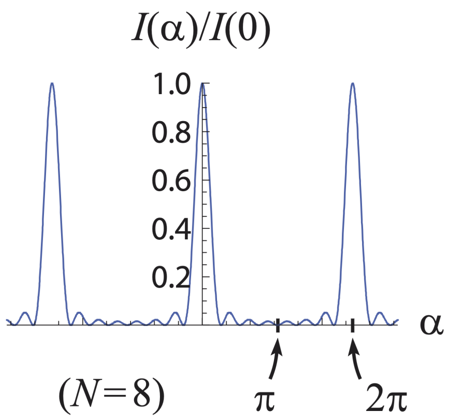


Figure 2: Interference pattern for $N = 8$ emitting sources. Credit: Morin

(a) Primary maxima of Eq. (3) occur when both numerator and denominator vanish simultaneously. Show that this requires $\alpha = 2m\pi$ for integer m. Hence show that the m-th primary maximum appears at angle θ_m satisfying

$$\sin\theta_m = \frac{m\lambda}{d}. \tag{4}$$

(b) Show that the width of each primary maxima (the full width, that is, the distance between adjacent zeros) is given by,

$$\Delta\alpha = \frac{4\pi}{N}\,, \tag{5}$$

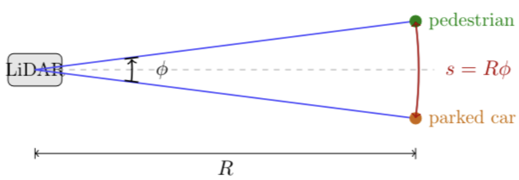
which for small angle approximation $\theta < 1$ translates to $\Delta\theta = 2\lambda/(Nd)$.

(c) The forward field-of-view of the car's sensor is from $-\pi/2 < \theta < \pi/2$. What is the condition between d and λ for there to be *at least* one other peak/beam on each side of the central maxima? (i.e. the $m = \pm 1$ beams must be physically realizable).

(d) Say the condition you derived in part (2.) above holds. In terms of the ratio, d/λ, how many such primary maxima peaks/beams would there be in the forward field of view?

(e) A typical LIDAR sensor on an autonomous car uses $\lambda = 900$ nm, and the sensors have a separation of $d = 30\,\mu$ m. How many beams does such a sensor have?

Our autonomous car now faces the following scene: directly ahead, a pedestrian and a parked car are standing at the *same* distance $R = 30$ m, separated by a transverse distance s:



(f) The two objects subtend an angular separation $\phi = s/R$ at the sensor. For the LIDAR be able to **resolve** them, that is, distinguish two separate objects rather than seeing them as one blurry blob, the beam width must be smaller than this separation:

$$\Delta\theta < \phi = \frac{s}{R}. \tag{6}$$

Write down the condition on N, d, λ, R, and s for the two objects to be resolvable.

(g) For $N = 8$, $d = 30$ μm, $\lambda = 900$ nm, and $R = 30$ m, compute the minimum resolvable separation $s_{\text{res}} \equiv R\,\Delta\theta$. For $s = 0.9$ m, is the sensor able to resolve the pedestrian and the car?

Resolving the two objects is necessary but not sufficient. Even if the beams are narrow enough to distinguish the two objects, the sensor must *simultaneously* illuminate both. There should be one object per primary beam. The relevant quantity is now the **angular spacing between adjacent primary maxima**, which sets a minimum object separation below which no two beams can simultaneously point at two different objects.

(h) From Eq. (4), show that the angular spacing between two adjacent primary maxima (orders m and $m+1$) near $\theta = 0$ is

$$\delta\theta_{\text{sep}} \approx \frac{\lambda}{d}. \tag{7}$$

For two objects at distance R to each be illuminated by a different primary beam, their separation must satisfy $s \geq s_{\text{sep}} \equiv R\,\Delta\theta_{\text{sep}}$. Compute s_{sep} for $N = 8$, $d = 30$ μm, $\lambda = 900$ nm, $R = 30$ m.

(i) Compute the ratio $s_{\text{sep}}/s_{\text{res}}$ and express it in terms of N. For the sensor to work, *both* conditions must be satisfied. Which one requires a larger minimum separation s and therefore dominates? Using your expression, explain in one sentence why increasing N does not help [2].

3 Michelson Inteferometer: Measuring Wavelength of Light

A **Michelson interferometer** is one of the most consequential optical instruments ever built. Invented by Albert Michelson in 1881, it was first used in the famous Michelson–Morley experiment (1887), which attempted to detect the motion of the Earth through the hypothetical "aether" — and famously found nothing, a null result that would later become a cornerstone of special relativity. More recently, a scaled-up descendant of the same idea — the LIGO detector — made history in 2015 by directly detecting gravitational waves for the first time, with arm lengths of 4 km and sensitivities to length changes of order 10^{-18} m, a thousand times smaller than a proton.

The operating principle is elegant: a single laser beam is split into two arms by a beamsplitter. Each arm travels to a mirror and reflects back; the two beams recombine and interfere. The output intensity depends entirely on the **path difference** ΔL between the two arms — constructive when $\Delta L = m\lambda$, destructive when $\Delta L = (m + \frac{1}{2})\lambda$. Any change in the length of either arm shifts ΔL and modulates the output — which is precisely how both the Michelson–Morley experiment and LIGO extract information from the world.

Open the simulation at `https://javalab.org/en/michelson_interferometer_en/` (works better in Chrome). Mirror 1 is movable; its position x is shown on a ruler in nm. Mirror 2 is fixed.

(a) **Path difference.** If Mirror 1 is displaced by Δx from its initial position, by how much does the path difference ΔL between the two arms change? (Think carefully about how many times the light traverses that displacement.) Hence show that one complete bright→dark→bright cycle corresponds to a mirror displacement of $\lambda/2$, and that observing m full cycles over a displacement Δx gives

$$\lambda = \frac{2\,\Delta x}{m}. \tag{8}$$

(b) **Simulation measurement.** Start at a bright maximum and record x_0. Drag Mirror 1 and count at least 5 full cycles, noting the final position x_f. Use Eq. (8) to estimate λ, then compare with the value displayed in the simulation. What is your percentage error?

[2]Think about it: $\Delta\theta = 2\lambda/Nd$ depends on N; $\delta\theta_{\text{sep}} = \lambda/d$ does not. More emitters sharpen each beam, but the *spacing* between beams is set entirely by d/λ, the geometry of the array, not its size.